Please read the instructions below before the exam starts, and listen carefully to any instructions given by the invigilators. You may read the question paper before the exam starts but you must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2023

EC120 2022/3 A 800
EC120 2021/2 A 850

Formula sheet with Statistical tables

Only Casio FX-83GTPLUS or FX-85TGPLUS or FX-83GTx or FX-85GTx Casio FX- 85MS calculators are permitted

UNIVERSITY OF READING

INTRODUCTORY QUANTITATIVE METHODS IN ECONOMICS AND BUSINESS 2 AND STUDY SKILLS (EC120)

Two hours

This paper is divided into **TWO SECTIONS.**

Answer **ALL** questions in Section A (40 marks); and
Answer **TWO** questions in Section B (60 marks).

SECTION A
Answer ALL questions

1. One organisation employs 42 male workers and 37 female workers. The average wage of male workers is £21.32 per hour with a standard deviation of 5.29. The average wage of female workers is £18.78 per hour with a standard deviation of 5.02.

   (a) Based on these information, can we conclude that in this organisation there is a gender wage gap in favour of men? Discuss your answer.

   (4 marks)

   (b) Based on these information, can we conclude that in the UK there is a gender wage gap in favour of men? Discuss your answer.

   (4 marks)

   (c) Would you be confident making conclusions about the UK given the information you have? Why yes/why not?

   (2 marks)

2. In the regressions below, Activity is the number of minutes spent on physical exercise on an average day, while Work is the number of hours worked on an average weekday. The standard errors are reported below each coefficient, and the sample size is 1,000 in each case.

   For each regression, interpret the slope coefficient.

   (a) Sample of 1,000 people aged 15-25
   Activity = 15.02 – 10.00 Work
   st. err. 8.10 9.23

   (5 marks)

   (b) Sample of 1,000 people aged 45-55
   Activity = 5.42 – 9.23 Work
   st. err. 6.70 3.01

   (5 marks)

3. Continuing with the example in Question 2: in the regressions below, Activity is the number of minutes spent on physical exercise on an average day, while Work is the number of hours worked on an average weekday. The standard errors are reported below each coefficient, and the sample size is 1,000 in each case.

   For each regression, compute and interpret the 95% confidence interval for the slope.

   (a) Sample of 1,000 people aged 15-25
   Activity = 15.02 – 10.00 Work
   st. err. 8.10 9.23

   (5 marks)

   (b) Sample of 1,000 people aged 45-55
   Activity = 5.42 – 9.23 Work
   st. err. 6.70 3.01

   (5 marks)

4. You are asked to design an experiment to test whether women work more hours than men. Discuss why you cannot use experiments to test this hypothesis.

   (10 marks)

SECTION B
Answer TWO questions
(30 marks each question)

5. For a sample of 10 students, the table below gives their level of self-esteem at the end of the academic year, measured on a scale from 1 to 10, and the weighted average of the marks their received in that academic year (on a scale from 0 to 100)

Student	Self-esteem	Marks Weighted Average
1	7.0	67.8
2	5.0	65.9
3	4.0	55.0
4	6.0	75.4
5	6.0	62.3
6	8.0	68.8
7	5.0	58.4
8	6.0	53.1
9	9.0	72.4
10	6.0	62.1

(a) Draw a scatterplot of these data. Describe the overall pattern in the scatterplot and any striking deviations from the pattern.

(5 marks)

(b) Compute (by hand) the sample mean and standard deviation of self-esteem and average marks.

(6 marks)

(c) Compute (by hand) the correlation between self-esteem and average marks. For full marks, show the steps in your computation.

(5 marks)

(d) Find a regression line which allows you to predict self-esteem given information on average mark.

(5 marks)

(e) Interpret the results of this regression.

(3 marks)

(f) What are out-of-sample predictions and why should we be careful when computing them?

(6 marks)

6. You have information on the number of Facebook friends and the number of close friends (who they meet regularly) for a sample of 451 young people (aged 20-25) and a sample of 43 older people (aged 60-65). For the two samples, the following table gives the averages number of Facebook friends and the standard deviations. The two sample sizes are in the row called “Observations”.

	Younger Aged 20-25		Older Aged 60-65	
	Mean	St. Dev.	Mean	St. Dev.
Facebook friends	250	132	89	75
Close friends	9	2	10	3
Observations	451		43	

(a) Describe the appropriate null and alternative hypotheses for testing whether:

(i) Younger and Older people have the same number of Facebook friends.

(ii) Younger and Older people have the same number of close friends

(4 marks)

(b) Carry out significance tests (using a 5% level of significance) to test these hypotheses (also state which critical value you are using and why).

(6 marks)

(c) Compute the 95% confidence interval for the difference in means for each hypothesis.

(6 marks)

(d) Interpret the results of the confidence intervals. What do we conclude?

(4 marks)

(e) Separately for the two samples, compute the 95% confidence interval around the mean for the number of Facebook friends (also state which critical value you are using and why).

(6 marks)

(f) Discuss why we normally prefer larger samples to smaller samples. [Hint: think about how the distribution of the sample mean depends on the sample size]

(4 marks)

7. A factory has 253 machines each producing the same items. Consider the following model relating the percentage of defective items produced by a machine ($Defective_i$) and the age of that machine in years (Age_i). Based on a sample of 47 machines, we obtain the following OLS estimates (standard errors in parenthesis):

$\widehat{Defective_i}$ = 0.01 + 0.09 Age_i
st. err.: (0.005) (0.045)

R^2 = 0.59

(a) Interpret the results of the model.

(6 marks)

(b) At the 5% level of statistical significance, test the hypothesis that the age of the machine has no effect on the percentage of defective items produced. What do you conclude?

(7 marks)

(c) Predict the proportion of defective items produced by a machine which is 25 years old. Discuss your answer.

(8 marks)

(d) Now imagine the owner of the factory decided to compute the regression above but using data on all 253 machines. The new OLS estimates are now:

$\widehat{Defective}_i$ = 0.015 + 0.088 Age_i
st. err.: (0.005) (0.044)

R^2 = 0.53

Based on these new results, can we conclude that age has a positive effect on the proportion of defective items? Discuss your answer and interpret the slope coefficient.

(9 marks)

(End of Question Paper)